

Interlinked UX deep research foundation for a serious transport simulation

First contact and menu authority

First impression experience

A serious simulation's first job is to establish an *emotional contract*: "this is a credible system you can trust, and it will reward thinking." Players coming from complex PC sims expect calm competence rather than hype — a sense of *authority*, *legibility*, and *stability* before the game even starts. That expectation is grounded in how quickly people form aesthetic judgements: visual appeal (and by extension, perceived quality/credibility) can be assessed extremely quickly, with research showing first-impression aesthetic judgements happening in tens of milliseconds. Gitte Lindgaard ¹ 's work is widely cited for the "–50ms" phenomenon in web contexts, which maps neatly onto games because the cognitive mechanism is the same: early perception drives a halo that colours what users think will happen next.

²

That halo matters because of the **aesthetic-usability effect**: attractive designs are often *perceived* as more usable, and users are more tolerant of minor friction if the interface looks "professional." That is not just vibes; it has a research base in HCI and applied ergonomics (for example, the influential "What is beautiful is usable" line of findings). Nir Tractinsky ³ 's work is a cornerstone here, and Nielsen Norman Group ⁴ summarises the practical implications for product design. ⁵

So: within the first minute, Interlinked should convey:

- **Composure over flash**: minimal visual noise, no "indie UI clutter", no overly stylised "faux-industrial" widgets that fight readability.
- **Systems legitimacy**: clear typographic hierarchy, consistent iconography, firm spacing rules, and interaction feedback that suggests engineering-grade care.
- **Predictability**: the player shouldn't be guessing whether a click is safe, destructive, modal, or irreversible.

You can see how strong simulation titles think about this in the way they treat "surface" UX as *part of the game design*. For example, Wube Software ⁶ publicly frames UI change as a trade-off between existing-player habit and new-player clarity — i.e., they optimise for long-term health and comprehension, not just legacy comfort. ⁷

What signals quality vs amateur design

In serious sims, "quality" is signalled less by ornament and more by **interaction integrity**:

1. **Coherent interaction grammar** (one mental model for selecting, placing, cancelling, confirming). When the grammar is consistent, complexity feels learnable.
2. **Information is where you expect it** (strong "information scent" and progressive disclosure rather than scavenger hunts). ⁸

3. **Errors are explainable and actionable** (warnings link to causes; the UI helps you recover quickly). ⁹
4. **Performance is respected** (waiting is handled intentionally, not as an awkward technical pause). ¹⁰

Common “amateur” signals in indie simulation UI are the inverse of those principles: inconsistent modes, unclear destructive actions, nested windows that occlude each other, tiny text with weak contrast, and “spreadsheet dumps” of data with no hierarchy. Game usability research has repeatedly found that interface friction and unclear feedback are recurring problem categories across genres, including strategy/simulation-heavy titles. ¹¹

What Interlinked should feel like in the first 60 seconds

Interlinked’s first minute should feel like stepping into a transport operations desk where everything is ready and calibrated:

- **Seconds 0–10:** Instant clarity: title, version, and a clean “Continue”/“Start” path. No noisy splash sequence; no “you’re in a launcher inside a launcher” feeling.
- **Seconds 10–30:** The menu presents a *clear promise*: sandbox/network-building is the primary experience; scenarios are optional training constraints; settings are discoverable and not buried.
- **Seconds 30–60:** If the player hits “Continue”, they sense momentum: the game is already thinking (streaming, sim warm-up) with an intentional transition and visible progress.

That “it’s already under control” feeling is exactly what progress-indicator research recommends: progress feedback reduces uncertainty and makes waits more tolerable, increasing willingness to wait. ¹⁰

Main menu design philosophy

Players coming from deep PC sims subconsciously expect the menu to behave like *an application launcher with save intelligence*, not like a console-style “press start” funnel. A gold-standard reference is how Factorio explicitly restructured its main menu around the most common player intent: “continue my last save” at the top, and then a clear separation of single-player versus multiplayer pathways. ¹²

That is a *menu philosophy*, not just a layout choice:

- **Intent-first hierarchy:** “Continue” is a primary action because it matches the dominant returning-player goal. ¹²
- **Mode separation by mental model:** single-player new/load grouped together; multiplayer hosting/browsing grouped together. ¹²
- **Content communication:** new players need descriptions of what “campaign/scenario/freeplay” actually mean; otherwise the menu becomes a jargon gate. ¹³

Transport-focused management games similarly treat sandbox/freeplay as the “default life” of the product. Transport Fever 2 ¹⁴’s documentation explicitly frames free game mode as the most relevant mode for most players, and its new-game flow is structured as a guided sequence (map selection → settings → advanced settings) with previews and seeds to support iteration and replayability. ¹⁵

Save management UX

In long-running sims, saves are not just files — they're *projects*. Players expect:

- A “Continue” that is honest (shows save name, last played, compatibility warnings).
- A “Load” screen that supports *recognition*, not recall: thumbnails/last viewport, key metadata, mod set, and a clear explanation when loading is blocked. Transport Fever 2's load UX shows save details and even the last viewport, and surfaces active mods for the selected save — reinforcing that a save has context, not just a timestamp. ¹⁵
- Options that don't destroy trust: if mods are missing, say so and help resolve (don't silently fail). ¹⁵

Continue vs load

“Continue” is an *action*, “Load” is a *decision space*. Mixing them is how menus get cluttered and slow. Factorio's rework rationale explicitly calls out the failure mode: important actions (like load) getting buried among rarely used options. ¹²

Settings placement

Players expect two-way settings:

- **Global settings** (graphics, audio, keybinds, UI scaling) from the main menu, before loading — because performance/readability problems often prevent comfortable play in the first place.
- **Session settings** (simulation speed, visual overlays, optional assists) in-game, because they are part of moment-to-moment control.

Transport Fever's manual reflects this expectation: some settings (like fullscreen resolution) are only available from the main menu, implying a clear split between global and in-session configuration. ¹⁶

Scenario vs sandbox separation

In serious sims, scenarios are typically framed as **constraint-based learning** (teach systems under pressure), while sandbox is **authoring + optimisation** (build a system over time). The menu should reflect that by:

- Making sandbox/freeplay obvious and first-class (not tucked inside nested “scenarios” unless clearly labelled).
- Presenting scenarios with clear descriptors: objective type, typical playtime, complexity, and what mechanics are highlighted. Factorio specifically notes the need for communicating what different scenarios are and what to expect. ¹²

Why Paradox-style philosophy matters here

Paradox grand strategy UX tends to assume: (1) the player will invest hundreds of hours, (2) the UI must support learning in-context, and (3) the UI is a knowledge system (tooltips, encyclopaedia, alert economy). Crusader Kings III ¹⁷ is an explicit example: its design describes “tooltips in tooltips,” reactive advice, and an in-game encyclopaedia that reduces the need to alt-tab to a wiki — a direct response to complexity and cognitive load. ¹⁸

For Interlinked, the key menu implication is: **don't treat onboarding as separate from the product**. The menu is part of learnability, and “authority” begins there. ¹⁹

Loading, transitions, and arrival

Loading into a save

Loading is not just waiting — it's a psychological moment where players decide whether the game feels *premium and dependable* or *technical and fragile*. Research on wait perception is consistent: progress indicators reduce uncertainty, increase tolerance, and improve satisfaction. ¹⁰

For serious sims, the risk is bigger than boredom: if loading feels unreliable or opaque, it undermines the player's confidence in the simulation as a whole (because the sim already asks for trust). ²⁰

Best-practice patterns are:

- **Always acknowledge** long operations quickly (even if the real work continues). We have classic response-time guidance showing that around ~1 second is where attention begins to drift without feedback, and progress indicators help reassure users the system is working. ²¹
- **Use truthful progress** where possible. Deceptive progress bars can backfire in expert audiences (and sim audiences skew expert). Research into progress-bar behaviour shows that progress presentation affects perceived waiting time and satisfaction; the implication is: design progress behaviour intentionally, not accidentally. ²²
- **Make load content useful to the player's next decision:** show world state hints, network "health," last session goals, or a short "what changed since last time" log — not random beginner tips that returning players will ignore. (If tips exist, they should be contextual: e.g., related to features present in the save/mod set.)

How to hide technical delays behind UX

The most effective "delay masking" is not visual filler; it is **giving the player a meaningful micro-task that aligns with their intent**.

A concrete design lesson comes from Factorio's blueprint library work: even in a highly technical system (blueprints network-synced in multiplayer), they aimed for the download time to complete "by the time the user moves the icon over its destination," i.e., hiding latency behind a natural interaction gesture rather than forcing a hard wait screen. ²³

In Interlinked, the analogue is letting pre-load steps feel like *prep* rather than delay:

- "Review last session alerts" while streaming.
- "Choose initial camera focus" (e.g., last problem area / HQ / busiest hub) as the world warms up.
- "Enable layers preset" (operations vs planning) before full sim unpauses.

Camera transitions and the 'arrival feeling'

Premium sim "arrival" is about *continuity*:

- Start from a known anchor (last viewport; last selected line). Transport Fever 2's load UX explicitly shows the last viewport before saving, reinforcing continuity and recognition. ¹⁵
- Fade in with a sense of the simulation breathing: time controls visible, pause state obvious, and the first on-screen cues are stable (no barrage of popups).

- If there's music, it should be static/tonal and low intrusion; if there's a stinger, it should feel like "systems online" not "action trailer".

This aligns with flow research broadly: players stay immersed when transitions preserve context and minimise disruptive cognitive switching. ²⁴

Best-practice definition for Interlinked

Loading should feel **intentional**:

- It communicates what is happening (streaming map, initialising simulation, validating mod set).
- It uses the time to re-orient the player toward their next move.
- It lands the player in a stable, paused (or clearly unpaused) state with immediate control and a visible next problem to tackle. ²⁵

In-game navigation architecture

Core in-game navigation feel

Serious transport/network games are map-first *systems design* tools disguised as games. Players build a mental model that looks like this:

- The **map is the workspace** (where cause and effect happens).
- UI chrome (bars/panels) exists to **change the map's state** or **interpret the map's state**.
- Everything else (stats, tutorials, encyclopaedia) exists to support decisions without pulling the player away from the workspace.

That mental model matches professional spatial tools. For example, QGIS ²⁶ describes the map canvas as the central display area, tightly tied to a layers panel that manages what is visible. The attribute table pattern (rows = features, columns = attributes) is a canonical "inspectable system" model. ²⁷

Transport Fever 2's UI is explicitly structured into persistent regions (data layers dropdown, warnings, pause/menu, finances, context help, construction/management tools, statistics tools, key values, speed/camera/date controls). That layout reveals a core sim UX truth: you need both **action tools** (build/manage) and **interpretation tools** (layers/stats/warnings) always within reach. ²⁸

Build mode vs view mode

Players expect a strong separation between:

- **Authoring actions** (build, bulldoze, assign, schedule).
- **Diagnostic views** (layers, heatmaps, overlays, inspectors).

But they also expect *fast switching* between them, because the transport gameplay loop is "diagnose → intervene → verify".

Transport Fever 2 demonstrates this duality directly: data layers are accessible via a dropdown, and warnings are a persistent "network health" surface, while construction tools and managers sit within the same overall toolbar. ²⁸

OpenTTD's UI shows a different, older-but-still-instructive approach: toolbars with specialised build modes, plus heavy use of modifiers and shortcuts (e.g., autorail as an efficient tool; Ctrl to remove or cycle signal types; dragging to place repeated signals). This is high power once learned, and importantly, the tool behaviour is *systematically consistent* across actions. ²⁹

Inspect vs modify

In complex sims, “inspect” needs to be safe and fast. The fastest way to break trust is to make the player fear clicking because they might accidentally alter the network.

Actionable pattern: **inspection is default; modification is explicit.**

- Hover/select shows info.
- A clear “edit/modify” affordance changes state.
- Destructive edits require friction (confirm, or undo).

This is consistent with game usability heuristics research: players need clear, immediate feedback and predictable controls, especially in complex interfaces. ¹¹

Planning vs operations

Transport games uniquely require players to think in two temporalities:

- **Planning time:** layout, capacity, topology, future-proofing.
- **Operations time:** delays, bunching, congestion, breakdowns, passenger overflow.

The UI should support two complementary working sets:

1. **Planning preset:** build tools + constraint previews + cost/coverage.
2. **Operations preset:** alerts + live KPIs + bottleneck diagnostics + dispatch tools.

A key note from professional transport planning software is that users frequently work through layers/filters and tabular inspections when diagnosing networks. For instance, PTV Group ³⁰ positions its flagship planning software as a macroscopic modelling tool used for network/demand analysis, and the surrounding documentation ecosystem emphasises layered analysis and exporting/inspecting data.

³¹

Interlinked can “gamify” this without losing seriousness by making those operations feel like *commanding a live system*, not surfing menus.

How players expect tools to behave

The most powerful interaction lesson from Factorio is how it treats tool state as *cursor state* — the player's hand is the mode indicator. The controls documentation formalises two critical behaviours: clear cursor (cancel), and pipette (pick the relevant thing instantly). Those are anti-friction moves that preserve flow in complex build environments. ³²

Tool expectations in a serious sim:

- **Cancel always works and is consistent** (no “ESC sometimes closes, sometimes doesn't” ambiguity).

- **Tool state is visible** (cursor/ghost/preview clearly indicates what will occur).
- **Repeating actions are optimised** (drag to place repeated elements; modifiers for batch operations; repeat last tool).
- **Expert power is additive**: shortcuts, not hidden complexity.

Reducing friction switching modes

The best mode-switching UX usually combines:

- **Sticky last-used tool** (return to prior tool context).
- **Quick toggles** between top-level categories (build/lines/vehicles/layers).
- **Contextual jump** from alert → location → relevant tool pre-selected.

Transport Fever 2's warnings design is a strong illustration: warnings provide context and let you select names to jump to where problems occurred — turning “what’s wrong?” into “take me there” rather than “good luck finding it.” ²⁸

How good simulation UI becomes “invisible”

UI invisibility in serious sims is not about hiding UI. It's about removing *extraneous cognitive load* so the player's working memory is spent on system reasoning rather than interface decoding.

Cognitive Load Theory (CLT) distinguishes intrinsic load (the inherent complexity of the task) from extraneous load (complexity caused by how information is presented). In serious transport sims, intrinsic load is necessarily high — the network is complex — so the UX must aggressively minimise extraneous load. John Sweller ³³ is foundational here, and later overviews stress designing to reduce extraneous load to avoid overload. ³⁴

Interlinked's navigation architecture should therefore prioritise:

- **Stable UI regions** (muscle memory).
- **Predictable layers and inspectors** (recognition).
- **Clear state changes** (no hidden modes).
- **Fast recovery** (undo, safe inspection defaults). ³⁵

Flow loop and agency

Player flow: what “feels good” in management games

The “addictive but not overwhelming” feeling in serious sims is best understood as a blend of:

- **Flow** (balanced challenge/skill, clear goals, immediate feedback). ³⁶
- **Competence + autonomy** (the player feels effective and self-directed). ³⁷

Penelope Sweetser ³⁸ and Peta Wyeth ³⁹ 's GameFlow model explicitly includes control, clear goals, and feedback as core elements of enjoyment — which maps directly onto transport/network management where the player is constantly diagnosing and optimising. ⁴⁰

Self-Determination Theory research applied to games (notably work by Richard M. Ryan ⁴¹ , C. Scott Rigby ⁴² , and Andrew K. Przybylski ⁴³) shows autonomy, competence, and relatedness predict

enjoyment and future play, and also links competence/autonomy perceptions to the intuitive nature of controls and a sense of immersion/presence. ⁴⁴

For Interlinked, “addictive” should not mean manipulative reward spam. It should mean:

- you spot a problem,
- you understand why it’s happening,
- you make a clean intervention,
- you see the network improve,
- and you *want* to chase the next constraint.

That’s engineering satisfaction.

What breaks flow in complex sims

The most common flow-breakers in complex management titles are not the systems — they’re UI friction and state ambiguity:

- Too many clicks for common actions.
- Needing to dig through menus to find a key metric.
- Hidden information (forcing guesswork).
- Confusing states (is the tool active? is it paused? did it apply?).
- Slow or unclear feedback loops. ⁴⁵

These issues are consistent with broader game usability heuristics literature, which repeatedly highlights clarity of goals, responsiveness, and consistency as core usability factors. ¹¹

Ideal gameplay rhythm and UX support

The loop you defined is essentially the prime loop of serious simulation:

Observe problem → investigate → plan → execute → observe result → iterate

Interlinked’s UX should support this rhythm by ensuring each phase has a “home” and minimal transition cost:

- **Observe:** persistent, glanceable KPIs + calm but noticeable alerts.
- **Investigate:** fast inspectors, contextual layers, “click-to-focus” from alerts.
- **Plan:** previews, constraints, and what-if affordances without irreversible commitment.
- **Execute:** fast build/modify tools; batch actions; undo safety.
- **Observe result:** immediate local feedback + longer-term trend feedback (did we actually fix the systemic cause?).

The Paradox-style alert economy is instructive here: in CK3’s dev diary text, alerts are framed as solvable items that disappear when you take relevant actions, and suggestions act as optional guidance/checklists — supporting both new and experienced players without permanently cluttering the interface. ¹⁸

Interlinked can adapt that philosophy to transport:

- Alerts for *actionable breakages* (line disconnected, deadlock, overcrowding).

- “Issues” or “Ops backlog” for *important but not urgent* optimisation opportunities.
- Suggestions as optional “next tasks” to prevent sandbox paralysis (“what should I optimise next?”). ⁴⁶

Emotional outcome of good flow support

When the UX supports the loop, players consistently report feeling:

- in control (they can act),
- competent (they understand cause/effect),
- curious (they want to test improvements),
- motivated to optimise (because results are legible and attributable). ⁴⁷

A useful framing here is classic flow literature (challenge-skill balance, clear goals, feedback), popularised for general audiences in *Flow: The Psychology of Optimal Experience* ⁴⁸ and adapted into game design language by creators such as Jenova Chen ⁴⁹ . ⁵⁰

Information design and complexity scaling

Information design: presenting large simulation data without overwhelm

Transport sims can drown players in numbers because transport is inherently multivariate (capacity, frequency, dwell time, transfers, congestion, reliability, cost, demand elasticity, etc.). The UX goal is not “show less data.” It’s: **show the right data at the right moment**, so the player *discovers* complexity as needed.

This is precisely where CLT is practical: you can’t reduce intrinsic complexity much, but you can reduce extraneous complexity by structuring how information is revealed and integrated. ³⁴

Progressive disclosure and progressive enabling

Jakob Nielsen ⁵¹ defines progressive disclosure as deferring advanced/rarely used features to secondary screens, making applications easier to learn and less error-prone. ⁵²

Importantly, serious sims often need **progressive enabling** as well (not only hiding UI, but pacing when systems become relevant). Factorio’s blueprint library discussion explicitly recognises that “blueprints” unlock a cluster of related systems at once (construction robots, planners, books, library), and explicitly aims to make that step smaller/easier for new players. ⁵³

A transport-sim analogue:

- Early game: basic lines/routes + simple demand.
- Mid game: signalling/priority + timetables + depot logic.
- Late game: network-level optimisation tools, advanced overlays, scenario editors.

Not by *locking content behind grind*, but by staging information so that when a tool appears, the player has a reason to use it.

Tooltips, inspectors, layers, and heatmaps

Complex sims succeed when “explanations are one gesture away.” CK3’s dev diary text is an explicit articulation of this: nested tooltips for highlighted concepts, modes to lock tooltips, reactive advice, and a searchable in-game encyclopaedia that reduces reliance on external wikis. ⁵⁴

Transport Fever 2 demonstrates a transport-specific interpretation layer:

- Data layers accessible from a persistent control.
- HUD icon filters to control map noise.
- A warning window that surfaces disruptions and supports navigating to problem locations. ²⁸

Cities-style sims show the power of multiple info views: Cities: Skylines is documented as having dozens of info views, and the traffic view highlights different vehicle categories and flow conditions. That supports a core player need: **see invisible dynamics** (traffic, demand, pollution, etc.) without requiring the player to infer from raw map geometry alone. ⁵⁵

Alerts and priority highlighting

Alert design is a major fork between “not overwhelming” and “constant irritation.” The best pattern is:

- Alerts mean *action is possible now*.
- Alerts link to the relevant location/tool.
- Non-urgent items go into a backlog/issue list.
- Alerts are configurable/learnable over time (new players want more; veterans want signal, not noise). ⁵⁶

This resembles “trust calibration” in automation contexts: users rely on systems when they understand how and when signals are meaningful. ²⁰

Information should be discovered, not dumped

Information Foraging Theory is a useful lens: users follow “information scent” — cues that predict value and cost of pursuing a path. ⁵⁷

For Interlinked, that means:

- Labels, icons, and previews should *predict* what the player will get before they click.
- Layers/overlays should have clear names and immediate visible effect (strong scent).
- Deep stats should be discoverable through a consistent “inspect” action, not buried in unrelated menus.

Complexity management in practice: hiding depth without removing it

A great “expert depth coexists with accessibility” example comes from OpenTTD’s settings culture: advanced options exist, but players can enable “expert (show all settings)” style depth when ready. A Steam discussion about enabling all signal types explicitly describes switching settings category to “Expert” to reveal complexity. ⁵⁸

Transport Fever 2 similarly includes advanced settings and cheats, but places them in a separate advanced dialog, with explicit consequences (e.g., achievements disabled when cheats enabled). ¹⁵

Factorio's UI design discussions show the same ethos: unify inventories/shortcuts, reduce ambiguous interactions (like unpredictable shift-click transfers), and guide players toward "proper shortcuts" rather than letting the UI generate chaos. ⁵⁹

Interlinked can translate these lessons into a principle: **make the "right way" the easiest way**, and make advanced power feel like an extension of the same rules, not a separate hidden game. ⁶⁰

Modern simulation UX and trust

Modern simulation UX principles in the 2020s

The 2020s shift in simulation UI is less about "minimalism everywhere" and more about **density with clarity**:

- Cleaner typography and spacing (less skeuomorphic framing).
- More contextual help (nested tooltips, searchable encyclopaedias).
- Better state signalling (hover previews, clear mode indicators).
- Higher baseline accessibility (UI scaling, contrast, colour-blind support).
- More intentional motion (microinteractions used as feedback, not decoration). ⁶¹

A relevant reference from the "minimal but information-rich" side is Mini Metro's visual design discourse: the Game Developer coverage of its minimalistic design highlights how the visual system is part of usability — simplifying representation so players can reason about routes and constraints quickly. ⁶²

Information density balance

Modern sims tend to avoid permanent "HUD wallpaper." Instead they:

- Keep core KPIs glanceable.
- Let players toggle/stack layers.
- Rely on inspectors for deep dives.
- Use contextual highlighting rather than always-on labels. ⁶³

This aligns with progressive disclosure and information foraging: show what's needed now, keep deeper options accessible, and provide strong cues. ⁶⁴

Dark mode and typography trends

Modern "serious tool" aesthetics often default to dark UI, but readability research is nuanced. NN/g summarises evidence that, for users with normal vision, positive polarity (light mode) often supports better performance for reading-heavy tasks, while dark mode can feel easier in dim conditions and is desired by many users. ⁶⁵

For Interlinked (a long-session game), the implication is not "pick one." It's:

- Offer theme options (or at least brightness calibration).
- Prioritise contrast and legibility in both.
- Avoid low-contrast "cool grey on grey" that looks modern but reads poorly.

Baseline contrast guidance from W3C ⁶⁶ (WCAG) sets widely used minimum ratios (e.g., 4.5:1 for normal text under common conditions) and explains the rationale tied to visual acuity and contrast sensitivity. ⁶⁷

Player trust and simulation credibility

In transport sims, *trust* is not optional. Players are constantly accepting system outputs they cannot fully verify (pathfinding decisions, demand models, congestion behaviour). Trust research in automation is directly applicable: when systems are complex and users can't fully understand them, trust influences reliance; the design goal becomes **appropriate reliance** — not blind trust, not permanent skepticism. J.D. Lee ⁶⁸ and K.A. See ⁶⁹ are foundational here. ⁷⁰

In games, the “simulation credibility” version of this is:

- Players will accept complexity if they can *trace cause and effect*.
- Players will reject outputs that look like fake numbers or arbitrary rules.

CK3's dev diary text contains an interesting trust-building move: the encyclopaedia content is described as being automatically generated from script files so values update when mechanics change — signalling that the UI is aligned with the underlying system, not hand-waved flavour text. ¹⁸

Transport Fever 2 builds trust via actionable diagnostics: warnings describe failures (no path, deadlock, station mismatch) and let players jump to the problem. That says “the sim is knowable; your fixes will be grounded.” ²⁸

How Interlinked should communicate “this is a serious simulation”

Interlinked should communicate seriousness through *transparency rituals*:

- **Numbers mean something:** each KPI has a definition, time window, and relationship to mechanics.
- **Changes are attributable:** when performance improves or degrades, the UI helps explain why (new bottleneck, capacity shift, route change).
- **The game admits uncertainty** (where relevant): instead of pretending perfect prediction, show confidence ranges or qualitative states (“congested”, “unstable headways”, “transfer penalty high”) — and back them with inspectable factors.

That approach matches appropriate-reliance principles: trust is strengthened when the system's behaviour and limits are legible. ⁷¹

Ergonomics, anti-patterns, and Interlinked design laws

Long session ergonomics

A serious transport sim invites multi-hour sessions. Ergonomics is therefore core UX, not polish.

Key long-session factors with strong research anchoring:

- **Eye strain and readability:** theme options and contrast discipline matter. Dark mode may feel more comfortable in darker environments but can reduce reading performance for normal-

vision users in many tasks; the best approach is giving players control and ensuring contrast/typography holds up. ⁷²

- **Cognitive fatigue:** reduce extraneous cognitive load (unnecessary searching, ambiguous modes). ³⁴
- **Interaction efficiency:** serious sim players expect keyboard shortcuts and consistent toggles because repetition is guaranteed. OpenTTD documents extensive hotkeys, and Transport Fever 2's docs include dedicated controls references for UI toggles — reflecting the genre norm that efficiency tools are part of mastery. ⁷³
- **Reducing repetitive actions:** batch operations, repeat last action, presets, and “do this everywhere” tools. These are power features that prevent burnout in late-game play.

Common UX mistakes in simulation games

Below are the failure modes that most reliably make sims feel clunky, confusing, or amateur — and why they break the experience:

1. **Ambiguous modes** (player can't tell whether they're inspecting, building, or editing). This spikes extraneous cognitive load and causes costly errors. ³⁵
2. **Non-actionable alerts** (warnings that don't link to causes or fixes). They create notification fatigue and erode trust. ⁹
3. **Information dumps without hierarchy** (giant panels of metrics with no “what matters now”). This overwhelms working memory and blocks learning. ⁷⁴
4. **Buried core actions** (load game hidden among rarely used options; core tools behind multiple clicks). Even Factorio explicitly called out “load game lost among never-used options” as a menu failure worth redesigning. ¹²
5. **Inconsistent control grammar** (cancel sometimes works; shift-click behaviour unpredictable; toggles don't toggle). Factorio's UI writing explicitly targets ambiguity like “where did shift-click transfer go?”. ⁷
6. **UI that occludes itself** (stacking windows with poor layering rules). Players often describe this as “UI fighting me,” and it damages the sense of control in management titles. ⁷⁵
7. **Fake-feeling simulation outputs** (numbers that don't respond predictably to interventions). This collapses credibility and breaks the mastery fantasy. ⁷⁶

These failure modes are consistent with formal game usability research, which identifies predictable controls, clear feedback, and accessible information as recurring quality determinants. ⁷⁷

What players actually want this to feel like

Synthesising the psychology and genre references, the emotional design targets that best fit Interlinked's serious transport niche are:

- **Calm control:** the game is complex, but you never feel lost — you feel like a professional operator.
- **System mastery:** learning sticks because the UI teaches in context and outcomes are attributable. ⁷⁸
- **Engineering satisfaction:** interventions “click” because diagnostics are clear and tools are efficient. ⁴⁷
- **Strategic clarity:** you always know the next constraint worth addressing (alerts/issues/suggestions done right). ¹⁸
- **Creative optimisation:** the UI enables experimentation without punishing curiosity (undo safety, previews, sandbox tools). ⁷⁹

This sits directly on top of SDT (competence/autonomy) and flow conditions (clear goals, feedback, control). ⁴⁷

UX design principles Interlinked should follow

Actionable “UX laws” for a serious transport sim (written as enforceable rules):

1. Default interaction is always **safe inspection**, never silent modification. ⁸⁰
2. Every tool has one consistent **cancel** behaviour, and it always works. ⁸¹
3. Never require more than **one navigation step** from an alert to its cause. ²⁸
4. Alerts must be either **actionable now** or routed into a non-urgent backlog (“issues”). ¹⁸
5. Always show the player **what state they are in** (paused, building, editing, routing). ⁸²
6. The map is the workspace; panels must never obscure the primary decision area without a clear reason. ⁸³
7. Use progressive disclosure: advanced tools are visible but not intrusive, and are introduced when they become relevant. ⁸⁴
8. “Continue” is sacred: if the player can keep playing, the UI should get them back in fast. ¹²
9. “Load” is a decision space: show thumbnails/metadata/mod set so players recognise saves. ¹⁵
10. Never hide a core metric behind an unrelated menu; **metrics belong near decisions** that use them. ⁸⁵
11. Every number shown has a definition, a time window, and an explanation path. ⁸⁶
12. If a tool is repeated, it must be optimised for repetition (drag, batch, repeat last). ⁸⁷
13. Keyboard shortcuts must cover high-frequency actions and be discoverable (visible in tooltips/help). ⁸⁸
14. Layer toggles must have immediate visible effect and clear naming (strong information scent). ⁸⁹
15. Provide presets for different working sets (planning vs operations) to minimise mode thrash. ⁹⁰
16. Reduce extraneous cognitive load relentlessly; don’t “teach through suffering.” ⁹¹
17. UI changes must be evaluated against new-player comprehension, not only veteran habit. ⁷
18. Never let UI windows occlude each other without clear layering rules and easy dismissal. ⁷⁵
19. Waiting is always acknowledged; longer waits always show progress and meaning. ¹⁰
20. Accessibility isn’t optional (contrast, scaling, colour safety), because sessions are long. ⁹²
21. Suggestions are optional guidance, not nagging; veterans can turn them down. ¹⁹
22. Make expertise feel like amplification, not like discovering secret UI rules. ⁹³

Ideal UX personality for Interlinked

Interlinked’s UX personality should be: **transport control centre meets modern strategy platform**.

Not sterile “enterprise SaaS,” and not playful mobile minimalism — instead:

- **Operational:** the interface speaks in system status, constraints, and actions.
- **Confident:** wording is clear and concrete; the UI feels like it knows what matters.
- **Quietly powerful:** depth is always there, but never shoved in your face.
- **Trustworthy:** it explains itself; it does not gaslight with fake precision. ⁹⁴

If designing the UX from scratch: experiential flow

This is an experience flow (not specific UI element design), from first launch to long-term play:

- **Launch:** the game opens with instant legibility and calm authority (no clutter, no friction). The player feels invited into a serious tool-like space. ⁹⁵
- **Menu:** the menu offers clear intent paths: Continue first; New game second; scenarios framed as constraints/training; settings accessible immediately; saves treated as projects with context. ⁹⁶
- **Load:** loading feels like preparation and re-orientation: visible progress, contextual “last session” anchors, and a smooth arrival into a paused/stable state. ⁹⁷
- **First minute in-world:** the player can pan/zoom/inspect safely. The game surfaces one or two *meaningful* starting signals (network objective, current bottleneck, or a gentle suggestion), avoiding pop-up spam. ⁷⁸
- **First hour:** the UX teaches through the real loop: observe → investigate → plan → execute → verify. It introduces layers/diagnostics exactly when needed, so the player feels smarter each time, not more confused. ⁹⁸
- **Long-term play:** the interface becomes an extension of mastery. The player develops personal workflows (presets, shortcuts, operations views). Complexity grows, but *extraneous* complexity stays low because the interaction grammar never changes — only the sophistication of the player’s decisions does. ⁹⁹

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